

Date: 03/09/2025

Lab Practical #09:

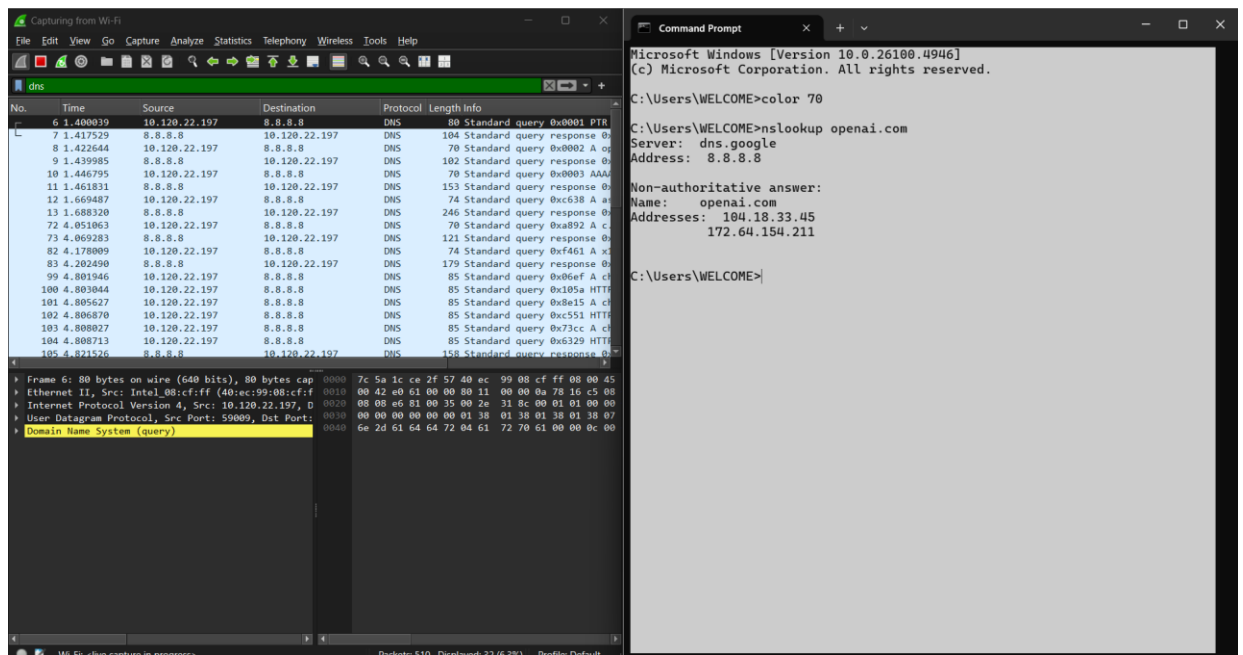
Study Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

Practical Assignment #09:

1. Explain usage of Wireshark tool.

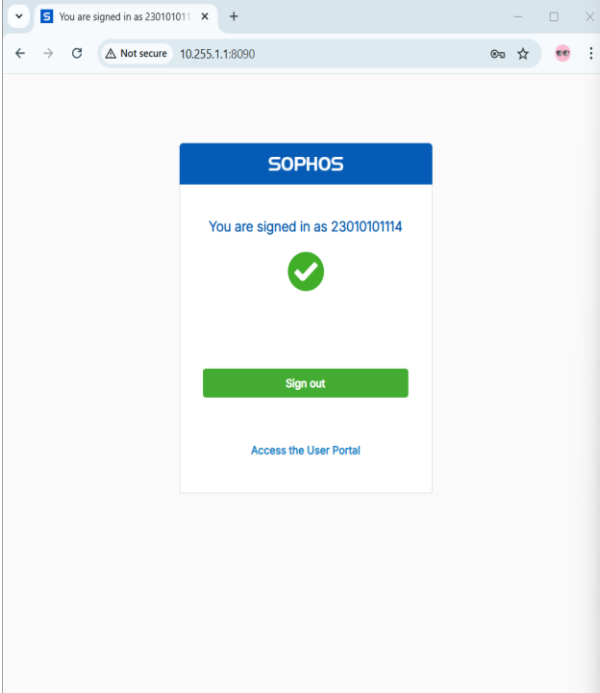
Wireshark is a network protocol analyzer used to capture and analyze network traffic in real time. It helps in troubleshooting network issues, monitoring performance, detecting security threats, and learning protocol behavior.

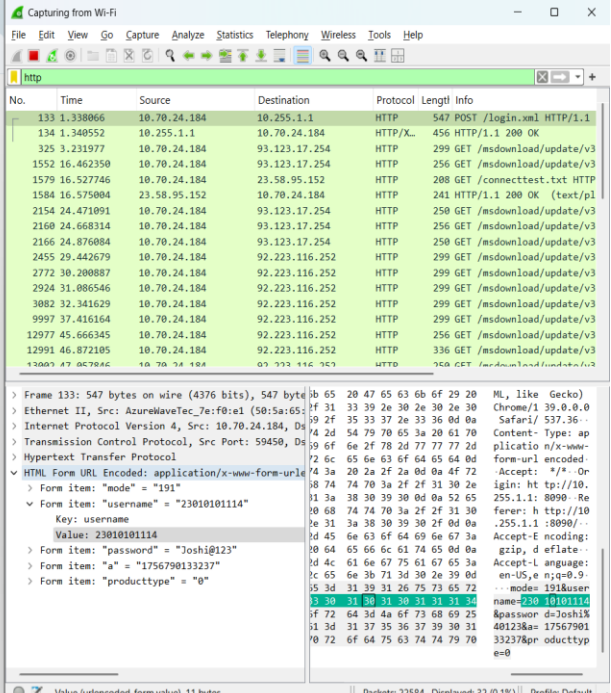
2. Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

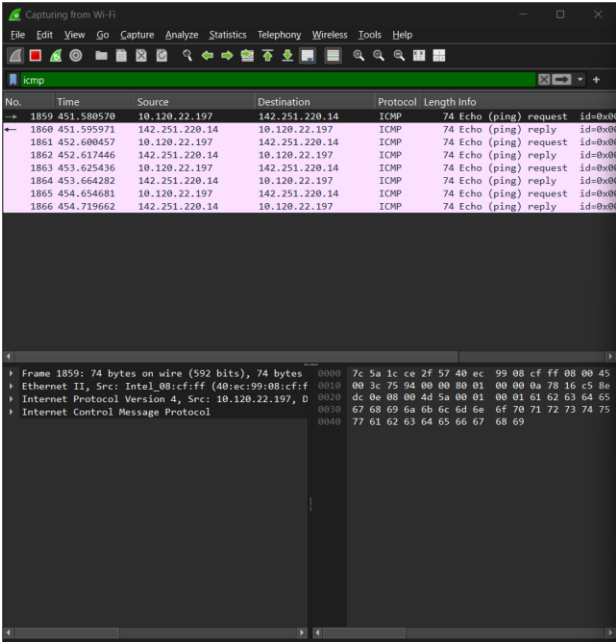


The screenshot displays two windows. The left window is Wireshark, showing a packet capture from Wi-Fi. The packet list shows a DNS query (No. 6) and a response (No. 7). The packet details pane shows the selected packet (No. 6) as a Domain Name System (query) packet. The packet bytes pane shows the raw data of the packet. The right window is a Command Prompt, showing the execution of the command 'nslookup openai.com', which returns the IP address 104.18.33.45.

Date: 03/09/2025











DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

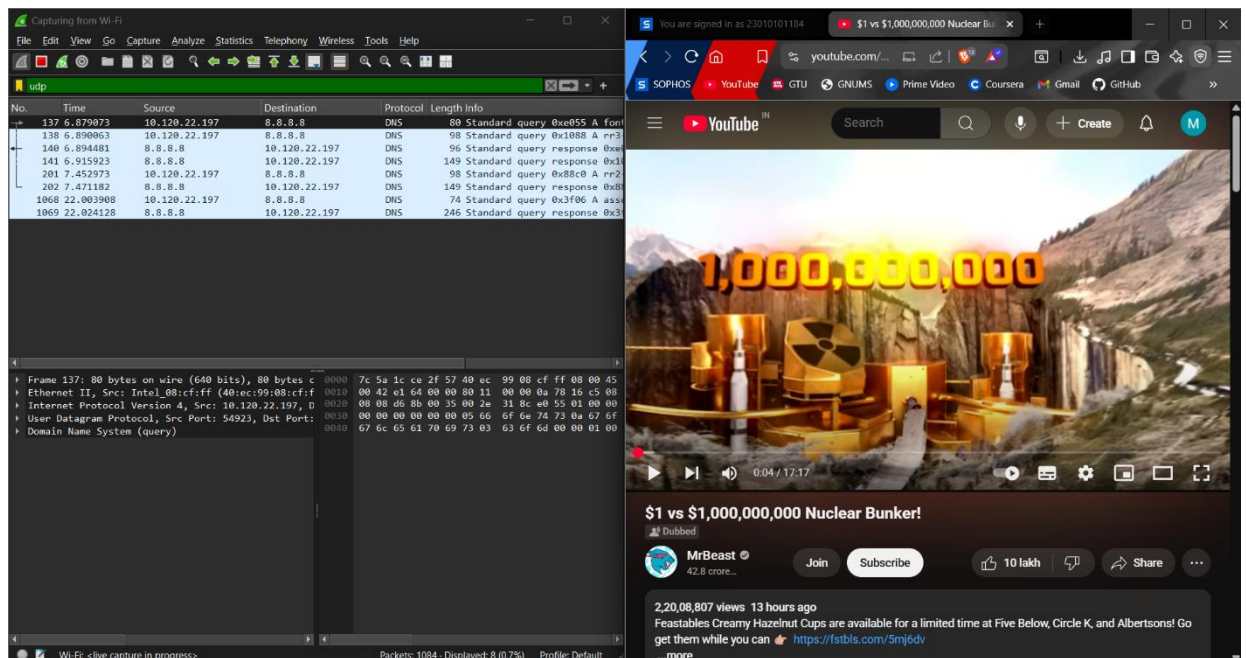
Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 03/09/2025

The image shows a web browser displaying the Darshan University Learning Management System (LMS) dashboard. The user is logged in as 230101011. The dashboard lists courses for Semester 5, including 2301CS402 - Design and Analysis of Algorithm and 2301CS501 - Computer Network. To the right, a Wireshark packet capture window is open, showing a list of captured packets. The selected packet is a TCP segment from 150.171.22.12 to 10.70.24.184, port 443, with a length of 1514 bytes. The packet details show it is a TCP segment with a sequence number of 443 and a destination port of 443.

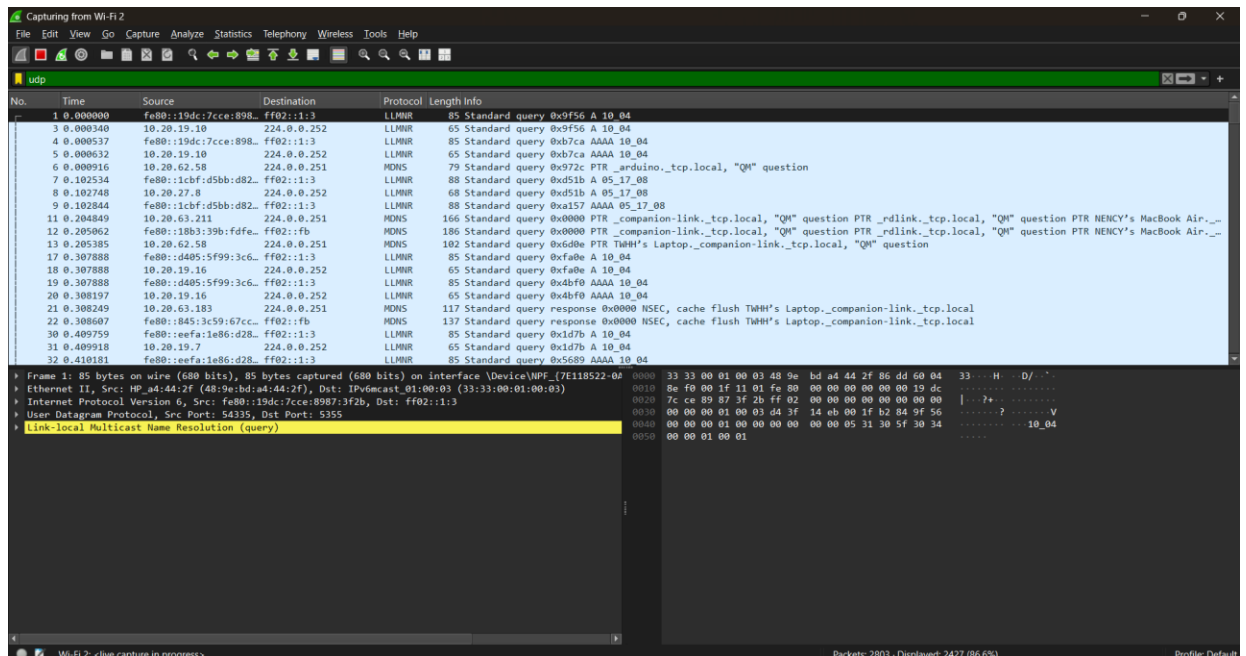
The image shows a detailed view of a Wireshark packet capture. The selected packet is a TCP segment from 150.171.22.12 to 10.70.24.184, port 443, with a length of 1514 bytes. The packet details show it is a TCP segment with a sequence number of 443 and a destination port of 443. The packet bytes are displayed in hexadecimal and ASCII format.

Date: 03/09/2025



The image shows a Wireshark packet capture of a DNS query. The packet list shows a query for '0x855 A 10_04' from source 10.120.22.197 to destination 8.8.8.8. The packet details show the query structure, including the question, answer, and authority sections. The packet bytes show the raw data of the query.

On the right, a YouTube video player is visible, showing a video titled '\$1 vs \$1,000,000,000 Nuclear Bunker!' by MrBeast. The video has 2,20,08,807 views and was uploaded 13 hours ago. The video content shows a nuclear bunker with a large '1,000,000,000' overlay.



The image shows a Wireshark packet capture of a DNS query. The packet list shows a query for '0x9F56 A 10_04' from source 10.20.19.10 to destination 224.0.0.252. The packet details show the query structure, including the question, answer, and authority sections. The packet bytes show the raw data of the query.

The packet details view is expanded, showing the 'Link-local Multicast Name Resolution (query)' section. The details show the query structure, including the question, answer, and authority sections. The packet bytes show the raw data of the query.